

Key: A Quarter of a Million Dollars and Nine Weeks

84-355 Democracy's Data | GOTV field experiments — retired | Instructor copy — do not distribute

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All Part 5 code is given to students. Grade the interpretation.

Every number below was run against the data file.

The lab in one sentence

Mobilization is the one part of campaigning that has been tested experimentally, and the experiments say the most *effective* tactic and the cheapest *per vote* are different tactics — while several of the things campaigns spend most on do nothing measurable at all.

The findings

```
e[, c("tactic", "contacts_per_vote", "cost_per_vote")]
```

```
#>
#> 1          Door-to-door canvassing          17          57
#> 2          Phone calls, volunteer           36          46
#> 3 Phone calls, commercial telemarketer      106         106
#> 4          Direct mail, nonpartisan        260         130
#> 5                Robocalls                425          64
```

```
g$tactic[!g$effective]
```

```
#> [1] "Direct mail, from a campaign" "Email"
#> [3] "Leaflets left on doors"
```

- **Door-to-door is the most effective** (1 additional vote per 17 conversations) and **volunteer phones are the cheapest per vote** (\$46 vs \$57). Not the same tactic — the whole lab hangs on this.

- **3 tactics have no measurable effect:** campaign direct mail, email, and leaflets left on doors. Not small effects — nothing distinguishable from zero.
- Spending the whole \$250,000 on the best-value tactic buys about **5,434 votes**. Wisconsin 2024 was decided by roughly 29,000.

The idea to teach

Two ideas, and the second is the harder one.

First: effectiveness and cost-efficiency are different axes, and campaigns optimise the wrong one when they confuse them.

Second, and this is the one worth the time: *“no significant effect” is a finding, not an absence of one.* Students read the blank cells as missing data. They are results — expensive, replicated, negative results. Getting them to feel the difference between “we don’t know” and “we know it doesn’t work” is the most transferable thing in this lab.

One housekeeping note before you teach it. `data/build-data.R` still says in its header that **four** tactics have no statistically significant effect, while the file it writes flags 3. The data is what this key and the brief report; the comment is what is stale. The likeliest candidate for the fourth is robocalls — see the anticipated questions — so reconcile the two before this lab comes out of retirement.

Option A — A different budget

Verified output

```
for (B in c(25000, 250000, 5000000)) {
  v <- floor(B / e$cost_per_vote)
  cat(sprintf("$%9s -> best %s, %s votes | order: %s\n", format(B, big.mark=","),
    e$tactic[which.max(v)], format(max(v), big.mark=","),
    paste(substr(e$tactic[order(-v)], 1, 12), collapse=" > ")))
}
```

```
#> $    25,000 -> best Phone calls, volunteer, 543 votes | order: Phone calls, > Door-to-door > Robocalls
#> $   250,000 -> best Phone calls, volunteer, 5,434 votes | order: Phone calls, > Door-to-door > Robocalls
#> $5,000,000 -> best Phone calls, volunteer, 108,695 votes | order: Phone calls, > Door-to-door > Robocalls
```

budget	votes
\$25,000	543
\$250,000	5,434
\$5,000,000	108,695

The ranking is identical at every budget, and votes scale linearly.

What a good answer says

3/4: Reports that the numbers scale and the best tactic does not change.

4/4: Sees why that is suspicious. **A model in which money converts to votes at a fixed rate has no diminishing returns in it** — but a \$5M program cannot buy 108,695 votes at \$46 each, because it would run out of volunteers, run out of doors worth knocking, and start contacting people who were always going to vote. The linearity is an artifact of using an average cost as if it were a constant.

4/4, best: turns that into the right conclusion about money in politics. This lab looks like it says money buys votes at a known price; what it actually says is that *the first* \$250,000 is efficient. The question “does money buy elections” needs the marginal, not the average, and this table has only the average. A student who says “I would need the experiments’ dose-response curves, not their headline” is doing graduate-level thinking.

Option B — Volunteer supply

Verified output

```
for (h in c(1000, 2000, 4000, CROSS, 12000)) {
  vv <- floor(h*CPH/VOL$contacts_per_vote); cost <- vv*VOL$cost_per_vote
  left <- BUDGET - cost
  tot <- if (left > 0) vv + floor(left/CAN$cost_per_vote) else floor(BUDGET/VOL$cost_per_vote)
  cat(sprintf("%5d hrs -> %5d vol votes, $%7s left, TOTAL %5d %s\n",
    h, vv, format(max(0,left), big.mark=","), tot,
    ifelse(left <= 0, " <- budget binds", "")))
}
```

```
#> 1000 hrs -> 694 vol votes, $218,076 left, TOTAL 4519
#> 2000 hrs -> 1388 vol votes, $186,152 left, TOTAL 4653
#> 4000 hrs -> 2777 vol votes, $122,258 left, TOTAL 4921
#> 7827 hrs -> 5435 vol votes, $      0 left, TOTAL 5434 <- budget binds
#> 12000 hrs -> 8333 vol votes, $      0 left, TOTAL 5434 <- budget binds
```

volunteer hours	total votes	binding constraint
1,000	4,519	volunteers
2,000	4,653	volunteers
4,000	4,921	volunteers
~7,827	5,434	the crossover
12,000	5,434	money

For reference: all-canvassing is 4,385 votes; unconstrained all-volunteer is 5,434.

What a good answer says

3/4: Finds roughly where the crossover is and states it.

4/4: Says what to do on each side of it. **Below ~7,800 hours, an extra \$50,000 is worth less than fifty more volunteers**, because the cheap capacity is already exhausted and marginal money buys \$57 canvassing. Above it, more volunteers sit idle and money is the constraint. This is why field directors are measured on recruitment: they are managing the binding constraint, not the budget.

4/4, **best**: notices that at 2,000 hours the mixed strategy (4,653) beats all-canvassing (4,385) but falls well short of the unconstrained ideal (5,434) — so the *published* \$46/vote figure is unreachable for almost every real campaign. A number that is true in the aggregate and unattainable in practice is worth being suspicious of, and students have now met that pattern twice (the other is Session 4's vote target).

Option C — Why buy the things that do not work

What a good answer says

This one has no output. It is graded entirely on the quality of the reason.

3/4: Offers one plausible reason — usually name recognition or “reaching people the other tactics miss.”

4/4: Gives a reason that survives the objection, and names the evidence that would test it. The strong candidates:

- **Turnout is not the objective; votes for *you* are.** Every number here is a turnout effect. Mail may not raise turnout while still moving *choice* among people already voting. The table simply does not measure persuasion.
- **Mail and email are fundraising instruments.** A mail program that loses on turnout and returns \$1.40 per dollar in donations is not irrational.
- **Signalling to donors, endorsers and press** that the campaign is real.
- **List-building** for later cycles.

4/4, **best**: notices the asymmetry in the evidence itself. The negative findings are about *turnout*, measured against the voter file, which is public. Persuasion effects require survey data and are much harder and rarer to measure. So the tactics that look useless may just be the ones whose payoff is hardest to observe — and campaigns that keep buying them are not necessarily being stupid, they may be responding to evidence the table cannot show.

Do not accept “campaigns are irrational” or “consultants profit from it” as a complete answer. Both are partly true and neither is testable as stated. Push for what evidence would settle it.

Option D — Their own question

Good versions: cost per vote if you value a *net* vote (mobilising both sides equally is worthless); what happens if door-knocking contact rates are half what the experiments assumed.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is your code.
- **The move that earns a 4** is treating the averages as estimates with conditions attached rather than as prices.
- **Reward anybody who separates turnout from persuasion** without being prompted. It is the distinction the whole table rests on and it is not labelled.

Anticipated questions

- “*Does 1 vote per 17 mean 16 people didn’t vote?*” — No, and this is the most common misreading. It means turnout in the contacted group was higher by about one vote per 17 people **than in an otherwise identical group that was not contacted**. Most of the 17 were going to vote anyway. Draw the treatment/control picture on the board; it takes two minutes and prevents half the bad write-ups.
- “*Why is canvassing more effective but more expensive?*” — Because a conversation at a door is more persuasive than one on a phone, and it costs far more per conversation (\$3.35 vs \$1.28). Effectiveness is per *contact*; cost is per *vote*. Two denominators. Across all 5 effective tactics the two orderings are an exact reversal — the rank correlation between contacts per vote and cost per contact is -1, which is the brief’s Step 5.
- “*Are these numbers current?*” — Green and Gerber’s 5th edition, 2024. Costs move with wages and postage; the effect sizes have been broadly stable across editions, which is itself notable.
- “*Do robocalls really work?*” — Barely: 1 per 425, which is 25 times as many conversations as door-knocking needs for the same single vote. The row clears the cost comparison only because a robocall costs \$0.15, not because it is good. **Say the caveat out loud.** **effective** is a TRUE/FALSE flag standing in for a significance test, and the file carries no standard errors, so a positive finding and a practically negligible one are indistinguishable in this column. The brief’s provenance section works this through and uses robocalls as the case in point. Nothing shipped in **data/** lets you re-verify that flag against Green and Gerber; treat it as the row most likely to move.
- “*Is this legal / is this what campaigns actually do?*” — Yes and yes. The experiments were mostly run with real campaigns and nonpartisan groups.

Connections

- **Session 9 (polls, models, and markets)** — same underlying problem: an average estimated on one population applied to another. There it is non-response, here it is external validity.
- **Chapter 12’s three factors** — ability, motivation, opportunity. Ask which one canvassing acts on. (Motivation, mostly. Registration law acts on opportunity. They are not substitutes.)
- **Session 4’s vote-targeting lab** — targeting says *where*, this says *how*. Together they are a field plan.
- **Tuesday’s debate** (political equality) — if mobilization mostly moves people who were nearly voting anyway, it may widen participation gaps rather than close them. That is the sharpest version of the question and worth saving for the last ten minutes.